

Q.25 Find the centre and radius of the equation of circle $x^2 + y^2 - 4x - 6y - 3 = 0$.

Q.26 Evaluate $\frac{dy}{dx}$ if $y = x^x$

Q.27 Differentiate $y = x \cos x$ with respect to x .

Q.28 Evaluate $\int \frac{2x}{x^2 + 3} dx$ by substitution method.

Q.29 Evaluate $\int_1^2 (2x + 1) dx$

Q.30 Find the area bounded by the curve $y = x^2$, x -axis and the lines $x = 2$ to $x = 3$.

Q.31 Find the length of latus rectum, focus and vertex of the parabola $y^2 = 24x$

Q.32 Evaluate $\int x \sin x dx$ by parts method.

Q.33 Differentiate $y = \sin(\cos x)$ with respect to x .

Q.34 Prove that $\sin 105^\circ + \cos 105^\circ = \cos 45^\circ$

Q.35 Resolve into partial fraction $\frac{1}{x(x+1)}$.

Section-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Calculate the area of triangle using Hero's formula if length of the sides are 3cm, 4cm and 5 cm.

Q.37 Use Simpson's rule to evaluate $\int_1^7 \frac{1}{x} dx$ by taking six equal intervals.

Q.38 Find the equation of circle which passes through the following three points (5, 7), (6, 6) and (5, -6).

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Section-A

Note: Multiple Choice questions. All questions are compulsory. (10x1=10)

Q.1 The value of 4P_2 is

- (a) 2 (b) 6
(c) 4 (d) 12

Q.2 The common ratio of the G.P. 3, 6, 12, 24, ... is

- (a) 2 (b) 3
(c) 6 (d) 10

Q.3 The value of $\sin 90^\circ$ is

- (a) 1 (b) 3
(c) -1 (d) 0

Q.4 The point (-1, 4) lies in which quadrant.

- (a) I (b) II
(c) III (d) IV

Q.5 The number of terms in the Binomial expansion of $(x+y)^{-2}$ is

- (a) 8 (b) 9
(c) 10 (d) ∞

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Q.6 The value of $\lim_{x \rightarrow 0} \frac{\sin x}{x}$ is

- (a) $\log_e a$ (b) 4
(c) 0 (d) 1

Q.7 $\int \cos x \, dx =$

- (a) $\sin x + c$ (b) $-\sin x + c$
(c) $\cos x + c$ (d) $-\cos x + c$

Q.8 The degree of the differential equation

$$\left(\frac{dy}{dx}\right)^3 + 2y \frac{dy}{dx} = \sin x$$

- (a) 0 (b) 1
(c) 2 (d) 3

Q.9 $\frac{d}{dx}(x^2) =$ _____

- (a) x (b) 1
(c) -1 (d) $2x$

Q.10 The value of $\int_0^1 e^x dx$ is

- (a) $e-1$ (b) 1
(c) 0 (d) $e+1$

Section-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Find the equation of circle with centre (h, k) and radius r.

Q.12 Find the 8th term of the AP: 1, 7, 13,

Q.13 The value of 5C_3 is _____

Q.14 The value of $\sin 30^\circ$ is _____

Q.15 Convert the degree measure 60° in to radian measure.

Q.16 $\sin (A+B) =$ _____

Q.17 Find the distance between the points A(3, 1) and B(6, 5).

Q.18 $\lim_{x \rightarrow 1} (2x + 5) =$ _____

Q.19 $\frac{d}{dx}(3x + 1) =$ _____

Q.20 $\int (x^2 + 1) dx =$ _____

Section-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Find the centroid of a triangle having coordinate of vertices as (2, 4), (4, 5) and (3, 3).

Q.22 Evaluate $\sin 75^\circ$

Q.23 Evaluate $\cos 40 \cos 20 - \sin 40 \sin 20$

Q.24 Find the equation of straight line passing through (2, 3) and making angle 45° with x axis.